

Lineage Permutation Groups and Complete Ordinary-Lane Survivor Frontiers

Guarded Relocation and Phase-Coset Boundaries

WuJun Chen

Independent Researcher

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This paper (Paper XXXVII of the RIME program) classifies intrinsic raw reachability and survivor incidence for five source lineages filling an ordinary lane. It makes no typed-transfer, projectability, settlement, or reset-bound claim.

Abstract

Problem. A fused-pair spectrum does not determine where the three surviving source lineages are placed. For branches that permute arithmetic lanes, an abstract permutation group also does not by itself establish that its elements are realizable by collision-free paths.

Approach. Fix normalized returns $\phi_r^{(a)} = \varepsilon_{\Delta} p^r a$ with $n = 5g$, $\Delta = g$, and $g \geq 2$. Five labelled lineages fill an ordinary five-point lane. The branch stabilizes the lane partition but may act by arbitrary permutations within ordinary lanes. All return labels are available, and paths have no length budget. Actual guarded words realize pure lane relocation and every local permutation generator.

Results. If c is the five-position phase cycle and σ_j are the ordinary-lane restrictions, then $K_a = \langle c, \sigma_1, \dots, \sigma_{g-1} \rangle \leq S_5$ gives exactly the reachable injections: one copy of K_a on each ordinary lane. The phase state set is the lane set times the right-coset set $H \backslash K_a = \{H\kappa\}$, where $H = \langle c \rangle$; deterministic labelled phase descent requires a separate normalizer condition. Every terminal placement comes from an element of the same K_a , and restricting that element to the three survivors gives the complete frontier. Symbolic matched families distinguish two losses of information: F_{20} and S_5 agree on unsigned labelled mobility, local parity, and fused-pair coverage but have two versus six survivor placements per pair; A_5 and S_5 have different reachable injection sets but identical complete survivor spectra.

Boundary. The classification concerns full ordinary-lane occupancy and unbudgeted raw paths. It does not classify partial occupancy, prove a universal minimal quotient, or supply an existing-semantics typed bridge.

Keywords. circular automata; guarded returns; packet lineages; permutation groups; phase cosets; survivor incidence

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/(5g)\mathbb{Z}, \Delta = g$	circular carrier and lane spacing
$p(q) = q + 1$	fixed full cycle
$E = Q \setminus \{0\}$	normalized section carrier
ε_Δ	collapse $0 \mapsto \Delta$, fixing all other points
$K_\Delta = \{0, \Delta\}$	terminal collision pair
$C_j, 1 \leq j < g$	ordinary five-point lanes
$C_*, \widehat{C}_*$	punctured and full kernel lanes
a, π, σ_j	branch, lane permutation, and local restrictions
$T_0 = \{q_0, \dots, q_4\}$	source-lineage labels in positive source order
$c(t) = t + 1$ on $\mathbb{Z}/5\mathbb{Z}$	phase cycle
K_a	five-position lineage permutation group
$\lambda_{j,\kappa}$	placement $q_i \mapsto j + \kappa(i)\Delta$
$\mathcal{L}_a(\lambda_0)$	injections reachable by actual guarded paths
$H \backslash K_a$	right cosets $H\kappa$, not a quotient group
$\mathcal{M}_a(\lambda_0)$	actual pre-collapse terminal placements
$K_a(F)$	terminal permutations sending the indices of F to $\{0, 1\}$
$\mathfrak{S}_a(\lambda_0; F)$	complete source-addressed survivor spectrum
$t_{01} = (0 \ 1)$	swap of the two terminal kernel indices

1 Introduction

A strict rank-five exit determines both a fused source pair and the placements of the three remaining lineages. The latter data are not recoverable from pair-level reachability alone. This paper asks for the complete raw survivor frontier when five source lineages fill an ordinary arithmetic lane and the branch restrictions are arbitrary permutations.

The distinction between group generation and actual paths is central. Knowing that several permutations generate a subgroup does not show that each generator, its inverse, and their composites satisfy the guards of the dynamical system. Conversely, knowing an existential coarse edge at each stage does not identify a single representative that can traverse the whole path. Source-addressed incidence is needed when such representative control is unavailable [5].

In the regime considered here, full ordinary-lane occupancy supplies uniform guards. The punctured lane has four positions and cannot receive five distinct lineages. A positive word realizes the inverse of the globally enabled branch return. Combining that word with a return label gives pure relocation between any two ordinary lanes. Relocation then turns each local branch restriction into an actual guarded loop at any current lane. These constructions realize every element of a single five-position permutation group, without changing representatives between independently chosen edge witnesses.

Paper XXXVI treats locally dihedral restrictions by retaining the lane and source-relative orientation [4]. Here orientation is replaced by a full permutation parameter. The resulting phase-coset set specializes to one cyclic state or two dihedral states per lane. For arbitrary local permutations, however, that state-set parameterization need not support deterministic labelled updates. The exact normalizer condition is stated separately.

The main results proceed in three steps.

1. Actual relocation and local loops give the complete reachable-injection classification, including a constructive reverse inclusion.
2. The same group gives exactly the actual terminal placements; its restriction to the three survivors gives the complete frontier and projection multiplicity.

3. Two symbolic matched families show that coarse mobility data can lose survivor information, while full reachable-state information can retain distinctions that survivor projection forgets.

The branch-normalized return law is inherited from Papers XXX and XXXI [2, 6]. All full-lane path constructions and terminal converses used below are proved here. No finite census is a premise of the classification.

1.1 Related Work and Novelty Boundary

Monotonic and cyclic-order-preserving automata have an established synchronization theory [7]. The present setting allows ordinary-lane restrictions that need not preserve cyclic order. Its question is not a shortest reset sequence but the exact source-labelled boundary of a guarded five-lineage execution.

Almost-group automata with one defect-one letter and permutation letters are a close ambient class. Casas Torres studies conditions for complete reachability, including the case of transitive imprimitive permutation groups [1]. The invariant here retains individual source lineages and requires injectivity on their complete support throughout the internal path. Support-image reachability and complete survivor incidence are different observables.

Dynamic-constraint synchronization concerns restrictions on the order in which a word traverses states [8]. The guard here is injectivity on a specified five-point placement at every internal return, followed by an actual terminal collision. The results below do not give a general algorithm or complexity classification for constrained synchronization.

Within the raw frontier line, Paper XXXV classifies arbitrary-branch one-lane survivor incidence by reachable cyclic orders and terminal-phase collapse [3]. Paper XXXVI supplies a signed mobility quotient and survivor theorem for multiple full ordinary lanes under locally dihedral restrictions [4]. This paper removes that local restriction in the same full-lane regime, proves actual relocation and local generator realization, and identifies the terminal restriction map for arbitrary ordinary-lane permutations.

Permutation-group generation, coset sets, and orbit-stabilizer counting are standard background, not novelty claims. The contribution is their exact realization by the declared guarded words and the resulting same-witness terminal classification. The related-work comparison is selective; no exhaustive priority or universal minimality assertion is made.

2 Normalized Full-Lane Dynamics

2.1 Carrier, Branch, and Source

Throughout,

$$n = 5g, \quad \Delta = g, \quad g \geq 2, \quad Q = \mathbb{Z}/n\mathbb{Z}, \quad E = Q \setminus \{0\}. \quad (2.1)$$

Let $p(q) = q + 1$ and define

$$\varepsilon_{\Delta}(q) = \begin{cases} \Delta, & q = 0, \\ q, & q \neq 0. \end{cases} \quad K_{\Delta} = \{0, \Delta\}. \quad (2.2)$$

The lane partition of E consists of

$$\begin{aligned} C_j &= \{j + t\Delta : t \in \mathbb{Z}/5\mathbb{Z}\}, \quad 1 \leq j < g, \\ C_* &= \{\Delta, 2\Delta, 3\Delta, 4\Delta\}. \end{aligned} \quad (2.3)$$

Its full kernel lane in Q is $\widehat{C}_* = C_* \cup \{0\}$. Assume

$$a \in \text{Stab}(\{C_*, C_1, \dots, C_{g-1}\}) \leq \text{Sym}(E). \quad (2.4)$$

The unique four-point lane is fixed setwise. There are a permutation π of $\{1, \dots, g-1\}$ and arbitrary $\sigma_j \in S_5$ such that

$$a(j + t\Delta) = \pi(j) + \sigma_j(t)\Delta. \quad (2.5)$$

Here $S_5 = \text{Sym}(\mathbb{Z}/5\mathbb{Z})$ acts on the five within-lane indices. No condition is imposed on $a|_{C_*}$. Fix distinct source labels $T_0 = \{q_0, \dots, q_4\}$ and one ordinary source lane $C_{\text{src}} = C_{j_{\text{src}}}$. Set

$$\lambda_0(q_i) = j_{\text{src}} + i\Delta, \quad \lambda_{j,\kappa}(q_i) = j + \kappa(i)\Delta \quad (\kappa \in S_5). \quad (2.6)$$

The initial enumeration gives the positive cyclic order of the source lineages. A lineage may denote any fixed nonempty source packet. Distinct source packets remain distinct labels before the strict exit; their atom counts are not read by the dynamics studied here.

2.2 Guarded Paths and Terminal Witnesses

For each $r \in Q$, the normalized return is

$$\phi_r^{(a)} = \varepsilon_\Delta \circ p^r \circ a. \quad (2.7)$$

Composition is right-to-left. A word lists its return labels in execution order, and acts on a placement by postcomposition of its coordinate map. For an injection $\lambda : T_0 \hookrightarrow E$, a return is enabled exactly when

$$K_\Delta \not\subseteq p^r a \lambda(T_0). \quad (2.8)$$

The unique nonsingleton fiber of ε_Δ is K_Δ , so (2.8) is equivalent to injectivity of $\phi_r^{(a)}\lambda$. Let $\mathcal{L}_a(\lambda_0)$ be the injections obtained by actual finite words whose every internal return is enabled, including the empty word. All labels are available, and no word-length budget is imposed.

A terminal witness consists of a reachable injection λ and a label u with $K_\Delta \subseteq p^u a \lambda(T_0)$. Its pre-collapse placement and strict incidence are

$$\mu = p^u a \lambda, \quad \beta = \varepsilon_\Delta \mu. \quad (2.9)$$

Write $\mathcal{M}_a(\lambda_0)$ for the actual pre-collapse terminal placements. For $F \subset T_0$, $|F| = 2$, put $R_0 = T_0 \setminus F$ and define

$$\mathfrak{S}_a(\lambda_0; F) = \left\{ \beta|_{R_0} : \begin{array}{l} \mu = p^u a \lambda \in \mathcal{M}_a(\lambda_0), \\ \mu(F) = K_\Delta \end{array} \right\}. \quad (2.10)$$

This is a set of source-addressed injections, not a count or mass profile. On R_0 , β agrees with μ . On F , both values of β equal Δ . Thus a survivor map together with the fixed F reconstructs the complete terminal incidence. Returning lineage labels to their fixed source packet blocks reconstructs the raw packet endpoint.

2.3 The Lineage Permutation Group

Let $c(t) = t + 1$ on $\mathbb{Z}/5\mathbb{Z}$ and define

$$K_a = \langle c, \sigma_1, \dots, \sigma_{g-1} \rangle \leq S_5, \quad H = \langle c \rangle \cong C_5. \quad (2.11)$$

K_a acts on within-lane indices, identified with the source labels through (2.6). It is not Paper XXX's physical return group on the normalized carrier, nor a group of typed transfer objects. Actual guarded realization of its elements is proved next, rather than assumed from this definition.

3 Actual Guarded Relocation and Local Generation

Lemma 3.1.

Every enabled return from a full ordinary lane ends in a full ordinary lane. If

$$\pi(j) + r = h + k\Delta \pmod{n}, \quad 1 \leq h < g, \quad (3.1)$$

then it is enabled on every injection filling C_j , and

$$\lambda_{j,\kappa} \xrightarrow{r} \lambda_{h,c^k\sigma_j\kappa}. \quad (3.2)$$

If $\pi(j) + r \equiv 0 \pmod{g}$, the return is strict rather than enabled.

Proof. The set $a(C_j)$ is $C_{\pi(j)}$. Rotation by p^r takes it to the full residue lane with residue $\pi(j) + r$ modulo g . If that residue is zero, the image is $\widehat{C}_*$ and contains both kernel points. Otherwise it is an ordinary lane C_h , which contains neither kernel point; ε_Δ fixes it. This decision depends on the support, not on its lineage ordering. Applying (2.5) coordinatewise gives (3.2).

Lemma 3.2.

For any ordinary lanes C_j, C_h , an actual guarded word $T_{j \rightarrow h}$ sends

$$j + t\Delta \mapsto h + t\Delta \quad (3.3)$$

on every representative filling C_j . Actual guarded loops also realize every phase c^t on every ordinary lane.

Proof. Let $m = \text{ord}(a)$. Since $a(E) = E$, $\phi_0^{(a)} = a$ is globally enabled. The word consisting of $m - 1$ zero labels implements a^{-1} ; when $m = 1$, this block is empty. Following it by the label $r = h - j$ gives

$$\begin{aligned} \phi_{h-j}^{(a)}(\phi_0^{(a)})^{m-1} \Big|_{C_j} &= \varepsilon_\Delta p^{h-j} \Big|_{C_j} \\ &= p^{h-j} \Big|_{C_j}. \end{aligned} \quad (3.4)$$

Every zero step sends a full ordinary lane to a full ordinary lane. The last step sends the representative back to C_h and is uniformly enabled by Lemma 3.1. Thus this is a word on the actual representative, not merely equality of formal coordinate maps.

The same inverse block followed by label Δ realizes $\varepsilon_\Delta p^\Delta$; on an ordinary lane it is the cycle c . Repeating this loop supplies all five phases, including inverse phases by positive powers.

Lemma 3.3.

Every $\kappa \in K_a$ is realized by a guarded loop at any ordinary lane.

Proof. At the current lane C_h , use the word

$$T_{h \rightarrow j} ; 0 ; T_{\pi(j) \rightarrow h}. \quad (3.5)$$

The semicolons denote execution order. Relocation first preserves the within-lane indices; the zero return then applies σ_j and moves to $C_{\pi(j)}$; the final relocation restores C_h without changing the new indices. Hence the loop realizes σ_j on every injection filling C_h . All three blocks are enabled on the actual predecessor produced by the previous block.

The loop returns to the same lane, so repetition realizes σ_j^{-1} by a positive power of its finite order. Lemma 3.2 supplies c and c^{-1} in the same way. Any expression for κ in these generators and inverses can therefore be executed as one concatenated guarded loop.

4 Complete Reachable Injections and Phase States

Theorem 4.1.

Under the hypotheses of Section 2,

$$\boxed{\mathcal{L}_a(\lambda_0) = \{\lambda_{j,\kappa} : 1 \leq j < g, \kappa \in K_a\}.} \quad (4.1)$$

In particular, $|\mathcal{L}_a(\lambda_0)| = (g-1)|K_a|$.

Proof. Initially $\lambda_0 = \lambda_{j_{\text{src}}, \text{id}}$. Lemma 3.1 shows that every enabled successor of $\lambda_{j,\kappa}$ is $\lambda_{h, c^k \sigma_j \kappa}$. If $\kappa \in K_a$, its successor permutation also belongs to K_a . Induction on the actual word gives the forward inclusion.

For the reverse inclusion, first use $T_{j_{\text{src}} \rightarrow j}$ to reach $\lambda_{j, \text{id}}$. Lemma 3.3 then realizes any desired $\kappa \in K_a$ at that lane. The resulting placement is $\lambda_{j,\kappa}$, and each step is enabled on its actual predecessor. Distinct (j, κ) give distinct injections.

Corollary 4.2.

Two reachable injections are phase equivalent when they occupy the same lane and differ by a positive Δ -rotation. Then

$$\begin{aligned} \mathcal{L}_a(\lambda_0) / \sim_{\text{ph}} &\cong \{1, \dots, g-1\} \times H \backslash K_a, \\ [\lambda_{j,\kappa}]_{\text{ph}} &\mapsto (j, H\kappa). \end{aligned} \quad (4.2)$$

Here $H \backslash K_a$ denotes the right-coset set $\{H\kappa : \kappa \in K_a\}$, not a quotient group. Its contribution to the state count is $|K_a|/5$.

Proof. A phase acts by $\lambda_{j,\kappa} \mapsto \lambda_{j, c^t \kappa}$. Thus equivalence is exactly equality of j and $H\kappa$. Theorem 4.1 supplies every (j, κ) , and Lemma 3.2 supplies every representative of its phase class by an actual loop. This proves the bijection and count.

Proposition 4.3.

Fix an enabled label from a source lane C_j . Its output phase class is independent of the input representative of each phase class if and only if

$$\sigma_j H \sigma_j^{-1} = H. \quad (4.3)$$

Consequently all internal labelled updates descend deterministically precisely when every ordinary-lane restriction normalizes H .

Proof. The target lane and exponent k in (3.1) do not depend on the representative. The phase representatives $c^t \kappa$ have output classes

$$H c^k \sigma_j c^t \kappa = H \sigma_j c^t \kappa. \quad (4.4)$$

These all equal $H \sigma_j \kappa$ exactly when $\sigma_j c^t \sigma_j^{-1} \in H$ for every t . This is inclusion of $\sigma_j H \sigma_j^{-1}$ in H , hence equality since both groups have order five. Each source lane has enabled labels to ordinary lanes, so the condition applies at every source.

Corollary 4.2 is therefore an exact compression of the reachable state set, not an unconditional deterministic transition congruence. When (4.3) fails, a quotient transition must retain representative incidence or be relation-valued; its existence cannot be inferred from the state-set bijection alone.

Corollary 4.4.

If every ordinary restriction is locally dihedral, then $K_a = C_5$ when all restrictions preserve orientation and $K_a = D_{10}$ when at least one reverses it. There are respectively one and two phase states per lane.

Proof. Rotations are powers of c . Any reflection together with c generates the dihedral group of the five-cycle, and all local restrictions belong to that group. The coset counts are 5/5 and 10/5. The two dihedral classes record source-relative positive and negative orientation.

Thus the signed state of Paper XXXVI is a specialization of the lane-by-phase-coset state, rather than a proposed minimal quotient for arbitrary local permutations.

5 Actual Terminal Placements and Complete Survivor Projection

Theorem 5.1.

For $\eta \in S_5$, define $\mu_\eta(q_i) = \eta(i)\Delta$. Then

$$\boxed{\mathcal{M}_a(\lambda_0) = \{\mu_\eta : \eta \in K_a\}.} \quad (5.1)$$

Every placement on the right is produced by one actual guarded prefix and one terminal occurrence.

Proof. By Theorem 4.1, a terminal source has the form $\lambda_{j,\kappa}$ with $\kappa \in K_a$. A terminal label u satisfies $\pi(j) + u \equiv 0 \pmod{g}$. Write $\pi(j) + u = k\Delta \pmod{n}$. Coordinatewise,

$$p^u a \lambda_{j,\kappa}(q_i) = (c^k \sigma_j \kappa)(i)\Delta. \quad (5.2)$$

Since $c^k \sigma_j \kappa \in K_a$, this gives the forward inclusion.

Conversely, fix $\eta \in K_a$ and any ordinary lane C_j . The permutation $\kappa = \sigma_j^{-1} \eta$ belongs to K_a , so Theorem 4.1 supplies an actual path to $\lambda_{j,\kappa}$. Take the terminal label $u = -\pi(j) \pmod{n}$. Then $k = 0$ in (5.2), and the pre-collapse placement is exactly μ_η . It fills $\widehat{C}_*$, so the occurrence is a strict exit. Also

$$p^{-u} \mu_\eta(T_0) = C_{\pi(j)} \subset E, \quad (5.3)$$

which verifies the section pullback on the same witness. No separate formal terminal placement is substituted for the one produced by the prefix.

Theorem 5.2.

For $F \subset T_0$, $|F| = 2$, let $I_F = \{i : q_i \in F\}$ and define

$$K_a(F) = \{\eta \in K_a : \eta(I_F) = \{0, 1\}\}. \quad (5.4)$$

Then

$$\boxed{\mathfrak{S}_a(\lambda_0; F) = \{q_i \mapsto \eta(i)\Delta \mid (q_i \in R_0) : \eta \in K_a(F)\}.} \quad (5.5)$$

In particular, F is hittable if and only if $K_a(F) \neq \emptyset$.

Proof. Every actual terminal placement is μ_η for an $\eta \in K_a$ by Theorem 5.1. It fuses F exactly when $\eta(I_F) = \{0, 1\}$. On all three survivors the collapse fixes the pre-collapse coordinates, which are $2\Delta, 3\Delta, 4\Delta$. Thus every actual survivor map belongs to the right-hand side.

For any $\eta \in K_a(F)$, use the prefix and terminal label constructed in the reverse inclusion of Theorem 5.1. That same occurrence both fuses F and places each survivor as in (5.5). Hence every displayed restriction is actual. Nonemptiness is equivalent to existence of such an occurrence.

The theorem classifies sets of maps. It does not replace one terminal witness by independent choices for the fused pair and the survivors. The resulting full incidence has value Δ on F and the displayed restriction on R_0 .

Corollary 5.3.

If $K_a(F) \neq \emptyset$, then

$$\begin{aligned} |\mathfrak{S}_a(\lambda_0; F)| &= \frac{|K_a(F)|}{1 + \mathbf{1}_{\{t_{01} \in K_a\}}} \\ &= \frac{|K_a|}{|\text{Orb}_{K_a}(I_F)|(1 + \mathbf{1}_{\{t_{01} \in K_a\}})}. \end{aligned} \tag{5.6}$$

Proof. Two permutations sending I_F to $\{0, 1\}$ and agreeing on the other three indices are either equal or differ by left multiplication by t_{01} . If $t_{01} \notin K_a$, the restriction map is injective. If $t_{01} \in K_a$, each fiber is the two-element set $\{\eta, t_{01}\eta\}$. This action is free and stays inside $K_a(F)$.

Since $\{0, 1\}$ lies in the orbit of I_F when $K_a(F)$ is nonempty, the set of permutations sending I_F to that pair is a coset of its setwise stabilizer. Orbit-stabilizer gives $|K_a(F)| = |K_a|/|\text{Orb}_{K_a}(I_F)|$.

For the cyclic and dihedral specializations, the hittable pairs are exactly the five source-consecutive pairs. Their survivor counts are one and two, respectively. The dihedral group contains no single transposition of two points: every reflection of a five-cycle exchanges two pairs of points. These cases recover the orientation distinction without extending it to arbitrary local branches.

6 Two Complementary Losses of Information

Fix $\pi = \text{id}$, let a be the identity on C_* and on every ordinary lane except C_1 , and vary only σ_1 . These branches exist for every $g \geq 2$. The source can be any full ordinary lane.

Proposition 6.1.

The following restrictions give the displayed lineage groups and survivor counts for each of the ten source pairs.

σ_1	K_a	$ K_a $	Survivors per pair
$t \mapsto 2t$	$F_{20} \cong C_5 \rtimes C_4 \cong \text{AGL}(1, 5)$	20	2
$(0 \ 1 \ 2)$	A_5	60	6
$(0 \ 1)$	S_5	120	6

Proof. Translation c and multiplication by 2 generate exactly the affine maps $t \mapsto \alpha t + b$ over $\mathbb{F}_5$: 2 generates $\mathbb{F}_5^\times$, and the translations give every b . These twenty maps act transitively on ordered distinct pairs and hence on the ten unordered pairs. No transposition belongs to this group. A nonidentity translation has no fixed point, and any affine map with $\alpha \neq 1$ has exactly one fixed point; a transposition has three.

The group generated by c and $(0 \ 1 \ 2)$ lies in A_5 . It contains $(1 \ 2 \ 3)$ by conjugation with c . Those two three-cycles fix 4 and generate a transitive subgroup on the other four points whose stabilizer of 0 contains $(1 \ 2 \ 3)$. Its order is at least $4 \cdot 3 = 12$, and at most 12 by evenness. Thus the full generated group has a point stabilizer of order at least 12; it is transitive by c , so its order is at least 60. It is therefore A_5 .

Conjugating $(0 \ 1)$ by powers of c gives all successive transpositions of the five-cycle. In particular, the adjacent transpositions along the path $0, 1, 2, 3, 4$ generate S_5 . Both A_5 and S_5 are transitive on unordered pairs: any permutation mapping one pair to another can have its parity corrected by swapping two points outside the target pair.

Theorem 5.2 and Corollary 5.3 now give the counts. The pair stabilizers have orders $20/10 = 2$, $60/10 = 6$, and $120/10 = 12$. The kernel swap is absent from the first two groups and present in the third.

Theorem 6.2.

For every $g \geq 2$, the branches with $\sigma_1(t) = 2t$ and $\sigma_1 = (0\ 1)$ have the same unsigned labelled lane-exit graph, the same local parity at every lane, and the same ten hittable source pairs. Nevertheless they have respectively two and six survivor maps for each pair.

Proof. Lemma 3.1 makes the unsigned internal graph depend only on π and the lane partition: label r is enabled at C_j precisely when $\pi(j) + r \not\equiv 0 \pmod{g}$, and its target lane is the unique ordinary residue lane with that residue. The terminal labels are exactly the complementary residues, again independent of σ_j . Both branches have $\pi = \text{id}$, so these labelled graphs agree.

Multiplication by 2 fixes 0 and cycles the four nonzero indices; it is odd, as is $(0\ 1)$. All other local restrictions in both branches are identity, so the local parities agree lane by lane. Proposition 6.1 gives all ten hittable pairs and the two distinct survivor counts. Therefore this combined coarse summary does not determine the complete survivor frontier.

Theorem 6.3.

For every $g \geq 2$, the branches with $\sigma_1 = (0\ 1\ 2)$ and $\sigma_1 = (0\ 1)$ have respectively $60(g-1)$ and $120(g-1)$ reachable injections. For every source pair F , their complete survivor sets nevertheless coincide: each is the set of all six bijections

$$R_0 \xrightarrow{\sim} \{2\Delta, 3\Delta, 4\Delta\}. \quad (6.1)$$

Proof. Proposition 6.1 identifies the groups as A_5 and S_5 , and Theorem 4.1 gives the injection counts. Fix any one of the six survivor bijections. It has exactly two extensions to a permutation sending I_F to $\{0, 1\}$, obtained by the two choices of kernel ordering. They differ by t_{01} and have opposite parity. Exactly one lies in A_5 , while both lie in S_5 . Theorem 5.2 realizes those extensions by actual terminal witnesses and forgets their kernel ordering on restriction. Hence both groups give every displayed survivor map and no others.

The two theorems express different directions of information loss. The first rejects a specific coarse summary. The second shows that the complete reachable injection set retains distinctions not visible in the survivor frontier. Neither identifies a universally minimal sufficient observable.

7 Scope and Open Boundaries

Full occupancy is a theorem hypothesis. The support of every reachable injection is an entire ordinary lane, so guard decisions are uniform and the punctured lane is inaccessible internally. With partial occupancy, holes can enable returns into the kernel residue lane, and the relocation and confinement proofs above no longer supply a classification.

All words are unbudgeted. The inverse branch block and repeated generator loops establish existence, not shortest paths or preservation of a length or credit allowance. A budgeted variant requires its own proof.

The general survivor theorem is parameterized by the embedded group $K_a \leq S_5$ and its restriction map. The paper does not assert an exhaustive five-stratum classification of subgroups containing H . The cyclic, dihedral, affine, alternating, and symmetric specializations are used only where their generation and restrictions have been justified. Group orders alone are not survivor-frontier data.

The raw classification supplies neither a complete typed source nor target-blind authorization, selector identity, declared transfer membership, or typed handoff. Those are separate compatibility obligations under existing semantics. No new forest semantics is a premise of the results here, and no projectable-origin or recursive-settlement conclusion is inferred.

8 Computational Artifacts

The paper-owned evidence package is under `experiments/paper37/` in the [RIME repository](#). Paths below are relative to this directory.

The control includes all 120 ordinary-lane restrictions at $g = 2$ with punctured identity, and 96 explicitly selected branches with 192 source cases at $g = 3$. Matched controls are separately replayed at $g = 2, 3, 4$. These are bounded consistency controls, not proofs of the all- g theorems.

Surface	Location	Role
Control	<code>ordinary_lane_audit.py</code> , <code>results/ordinary_lane_audit_v1.json</code>	bounded source-addressed replay
Lean	<code>lean/Paper37.lean</code> , <code>lean/formalization-manifest.json</code>	partial implication spine
Validation	<code>validation/validate_release.py</code>	source binding and finite replay
Evidence	<code>development-manifest.json</code>	exact 22-artifact nested closure
Release	<code>release-manifest.json</code>	PDF, source, and environment binding
Receipt	<code>results/</code>	downstream local closure check

The partial Lean spine formalizes word composition, conditional control reduction, phase-state equivalence and normalization, survivor restriction multiplicity, and the algebraic A_5/S_5 projection equality. The concrete guard, confinement, relocation words, and terminal-placement converse remain manuscript proofs and explicit inputs to the formal reduction.

The receipt records local closure verification and finite replay, not a fresh Lean compilation or independent mathematical validation.

9 Claim Status and Boundary

Claim	Status
Lemmas 3.1–3.3: uniform edges and actual relocation/generator words	proved for full ordinary lanes
Theorem 4.1: complete reachable injections	proved for all $g \geq 2$ in the declared regime
Corollary 4.2: lane-by-phase-coset state set	proved as a set bijection; no quotient-group claim
Proposition 4.3: deterministic labelled phase condition	proved under the exact local normalizer condition
Corollary 4.4: cyclic/dihedral recovery	proved specialization
Theorems 5.1–5.2 and Corollary 5.3: terminal and survivor classification	proved with actual witnesses and exact restriction multiplicity
Proposition 6.1 and Theorems 6.2–6.3: matched symbolic controls	proved for all $g \geq 2$; not finite certificates
Fixed $g = 2$ slice, selected $g = 3$ catalogue, and matched $g = 2, 3, 4$ replay	bounded consistency controls; not all- g proofs
Lean word/control, phase, and restriction spine	partial formalization; geometric inputs remain explicit
Partial occupancy and budgeted paths	open; not classified
Exhaustive subgroup taxonomy	not claimed
Existing-semantics typed alignment, projectability, settlement, reset bounds	outside the theorem surface

10 Conclusion

In the full ordinary-lane regime, actual guarded relocation words turn arbitrary local branch restrictions into controllable generators at every ordinary lane. Their group gives exactly the reachable source-labelled injections and the actual terminal placements. Restriction of a single terminal permutation then gives the complete survivor frontier.

Phase may be removed at the level of the reachable state set, but deterministic labelled phase descent requires normalization of the phase group by each local restriction. Survivor projection has its own loss of information: unsigned mobility, parity, and pair coverage can be too coarse, whereas different complete reachable state spaces can still have identical survivor frontiers. These are scoped raw statements, not typed recursive compatibility or reset estimates.

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